Order Sirenia

Manatees and dugongs may be the marine equivalent of terrestrial elephants. These large, aquatic herbivores give birth to well developed, singleton offspring who suckle from a pair of axillary mammae.

Order Sirenia, Family Dugongidae, Genus *Dugong*

Dugongs (*Dugong dugon* 250-1000 kg) and the extirpated Stellar's sea cow (*Hydrodamalis gigas* 4000 kg) are tropical and subtropical, aquatic grazers. The non-deciduate, epitheliochorial or hemochorial, villous, diffuse or zonary placenta implants into a bicornuate uterus (231, 4435, 6835, 6838, 6842, 7563, 7936, 10966, 10967, 10968). Conception can occur during lactation (6840). Neonates are furred, although the hair falls out during development (5758, 10158). Males have tusks, inguinal testes, a diffuse prostate and bulbourethrals, and large seminal vesicles (4417, 6837, 6839, 7936). Details of anatomy (6835, 6838), birth (5106, 6707a), and suckling (276) are available.

NEONATAL MASS		
Dugong dugon	20 kg; n1 stillborn	193
	20-35 kg	6840, 6842, 7936
NEONATAL SIZE		
Dugong dugon	100-130 cm	6840, 6842, 7936
	109-150 cm; small calves with solid food in stomachs	5106
	128 cm; n1 stillborn	193
	140 cm; n1 emb	5106
LITTER SIZE		
Dugong dugon	mode 1	2449
	mode 1 yg; twins recorded	6840, 6842
	mode 1; r1-2	10809
	1; 2 rare	10158
	1; 2 rare	5758
Hydrodamalis gigas	1 yg	4217
SEXUAL MATURITY		
Dugong dugon	f: 1st ovulate: 9-10 yr	6840
	m: est 9-10 yr	6840
	f/m: ~2 yr; 2.4 m; 248 kg	4586, 6842, 10242
	f/m: 9-9+ yr	6834
	f/m: 9-10 yr, sometimes 15-17 yr	10809
	f/m: 9-14 yr	10698
GESTATION		
Dugong dugon	11-12 mo	5758
	333 d	2449
	est ~1 yr	6840, 6842
	13-14 mo	7936
Hydrodamalis gigas	>1 yr	4217
LACTATION		
Dugong dugon	solid food: 3 mo	5106
	wean: ≥1.5 yr	6840
	with f: 18+ mo	10698
	with f: 1-2 yr	10809
INTERLITTER INTERVAL		
Dugong dugon	≥3 yr	10698
	3-6 yr	10809
	3-7 yr	6840, 7936
SEASONALITY		
Dugong dugon	spermatogenesis: May-Oct; n QLD, Australia	6834
	preg f: Feb-Mar; India, Sri Lanka	8499
	birth: Mar-June; nw Australia	6707a
	birth: Aug-Sept; n QLD, Australia	4586
	birth: Aug-Dec; n QLD, Australia	6840
	birth: Sept-Nov; n QLD, Australia	6834
	birth: Sept-Dec; QLD, Australia	10809
	birth: Oct-Apr; New Guinea	10809
	birth: peak Sept-Dec; n QLD, Australia	7936
Hydrodamalis gigas	birth: fall; Commander I, Bering Sea	4217

Order Sirenia, Family Trichechidae, Genus *Trichechus*

Manatees (*Trichechus inunguis*, *T. manatus*, *T. senegalensis*) are 150-650 kg, aquatic grazers from tropical and subtropical west Africa, the Amazon basin, and the New World Atlantic. The hemochorial, labyrinthine, zonary placenta implants into a bicornuate or bipartite uterus and may have a partly endotheliochorial accessory placenta (231, 1545, 7563, 7628, 11374, 11850). Females have 2 axillary, mammae (1474, 1545, 7498b). Males have seminal vesicles and abdominal testes (1545, 11374). Manatees do not have a pineal (8869). Details of anatomy (7498b), birth (578, 7498b), and milk composition (475, 8398) are available.

NEONATAL MASS		
Trichechus manatus	11-27 kg	5108
	~27 kg	5111
	28.6 kg; n1	578
	m: 31.8 kg; n1	11636
	34.3 kg; r27.7-36.3; n7	11636
	42.0 kg; n1	4454
NEONATAL SIZE		
Trichechus inunguis	73.6 cm; n1 small calf	5105
Trichechus manatus	99 cm; n1	578
	120 cm	9079
	m: 120 cm; n1	11636
Trichechus senegalensis	~100 cm	5107
	104.5 cm; r104-105; n2 small calves	5107
WEANING MASS		
Trichechus inunguis	65-70 kg; hand reared 1 yr after capture	875
LITTER SIZE		
Trichechus manatus	1 yg; n1 f	7500
	1 yg; n1 f	578
	1-2 yg; 2 rare	2211
	1; 2 occasional	5111
Trichechus senegalensis	1 yg	5107
SEXUAL MATURITY		
Trichechus inunguis	f/m: 3 yr	2211
Trichechus manatus	f: 3-5 yr, 2.6 m	4454
	m: 3-5 yr, 2.5 m	4454
	f/m: 4-4.5 yr	2211
GESTATION		
Trichechus inunguis	180-270 d	2211
	12-14 mo	10809
Trichechus manatus	5-6 mo	5111
	≥152 d	7498b
	180-270 d	2211
	~11 mo	1474
	385-400 d	5108
	~13 mo	4454
LACTATION		
Trichechus inunguis	wean: 1 yr after capture; hand reared	875
	wean: ~2 yr	2211
Trichechus manatus	solid food: 38 d	4454
	solid food: 3 mo	1545
	solid food: 4-5 mo	578
	wean: 1 yr	2211
	wean: 1-2 yr	4454
	nurse: 18 mo	578

INTERLITTER INTERVAL		
Trichechus manatus	3-3.5 yr	4454
SEASONALITY		
Trichechus inunguis	conceive/birth: June; Amazon R, S America	10809
	birth: Dec-July, 63% Feb-May; Amazon R, S America	872
Trichechus manatus	m fertile: yr rd; Atlantic	1545
	mate: Dec; Crystal R, w FL	4454
	birth: May, Aug, Nov; captive	7499
	birth: yr rd; e US, Mexico	1474
	birth: yr rd; Suriname	5111
	young found: yr rd; FL	7499